Wavetable Filter

User Manual

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Overview

Wavetable Filter is an audio effect plugin (VST3/CLAP) that uses wavetable frames as FIR filter kernels. Instead of using traditional filter designs, the frequency response is shaped by the spectral content of a wavetable — giving you complex, evolving filter shapes that aren't possible with conventional EQ or filter plugins.

The plugin supports two filtering modes:

- **Raw** — Direct time-domain convolution. Linear-phase response with pre-ringing around transients.
- **Phaseless** — STFT magnitude-only filtering. Preserves transient shape with no pre-ringing, at the cost of slight temporal smearing from the STFT overlap-add process.

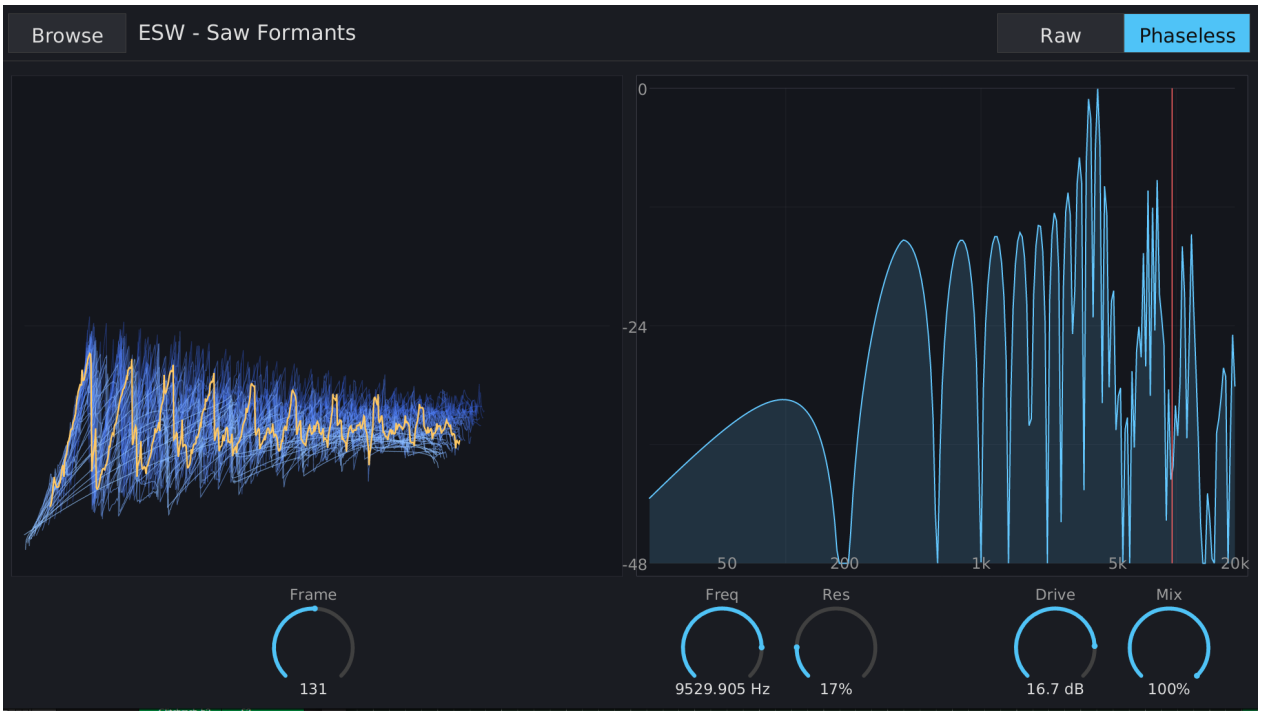

Figure 1: Wavetable Filter interface

Interface

Header

- **Wavetable Filter** — Plugin title.

- **+/- buttons** — Scale the UI from 100% to 300% in 25% increments. The scale is saved with your DAW project.

Wavetable Section

- **Wavetable** — Displays the name of the currently loaded wavetable. Hover to see the full file path.
- **Browse...** — Open a file dialog to load a `.wav` or `.wt` (Surge-compatible) wavetable file.
- **Mode** — Switch between **Raw** and **Phaseless** filtering modes. Click to select.

Visualizations

- **3D Wavetable View** (left) — Shows all frames of the loaded wavetable in an overhead perspective. The active frame is highlighted in orange. Click the view to toggle between the 3D overview and a 2D face-on view of the current interpolated frame.
- **Filter Response** (right) — Shows the frequency response of the current filter kernel on a log-frequency/dB scale. A faint amber shadow shows the input audio spectrum when audio is playing, making it easier to set the cutoff by eye. The red vertical line marks the cutoff frequency.

Controls

All controls support vertical drag (up = increase, down = decrease), shift+drag for fine adjustment, double-click or right-click to reset to default, and scroll wheel for stepped changes.

Frame Position

Selects which frame of the wavetable is used as the filter kernel. Adjacent frames are interpolated smoothly. Automating this parameter morphs the filter shape over time.

- Range: Frame 1 to the last frame in the wavetable
- Default: Frame 1

Frequency

Sets the cutoff frequency of the filter. The wavetable's harmonic content is resampled so that harmonic 24 of the wavetable maps to this frequency.

- Range: 20 Hz to 20,000 Hz (logarithmic)
- Default: 1,000 Hz

Resonance

Applies a comb-like emphasis at harmonic intervals of the cutoff frequency. At 0, the filter response is determined purely by the wavetable shape. Higher values create peaked, ringing resonances.

- Range: 0% to 100%
- Default: 0%

Gain

Output gain applied after the filter and dry/wet mix.

- Range: -20 dB to +20 dB
- Default: 0 dB

Mix

Blends between the dry (unfiltered) input and the wet (filtered) output.

- Range: 0% (fully dry) to 100% (fully wet)

- Default: 100%

Filter Modes

Raw Mode

Uses direct time-domain convolution with the wavetable frame as a 2048-tap FIR filter kernel. The convolution is SIMD-accelerated (16-wide f32 lanes).

Characteristics:

- Linear phase response — no phase distortion
- Symmetric pre-ringing around transients (characteristic of linear-phase FIR filters)
- Zero latency reported to the host
- Crossfades between kernels over ~20 ms when parameters change to avoid clicks

Phaseless Mode

Uses Short-Time Fourier Transform (STFT) with 50% overlap-add. The input signal is windowed, FFT'd, multiplied by the filter's magnitude spectrum, and inverse-FFT'd. Only the magnitudes are applied — the input signal's phase is preserved.

Characteristics:

- No pre-ringing — transient shape is preserved
- Slight temporal smearing from the STFT overlap-add (inherent to the technique)
- Introduces latency of 1024 samples (reported to the host for automatic delay compensation)
- Frequency resolution of ~23.4 Hz per bin at 48 kHz

Wavetable File Formats

WAV files (.wav)

Standard WAV audio files. The file is divided into equal-sized contiguous frames. The frame size is determined by the total sample count divided by the number of frames. Common frame sizes are 256, 512, 1024, or 2048 samples.

Surge Wavetable files (.wt)

Surge-compatible wavetable format with frame metadata embedded in the file header. These files explicitly encode the frame size and count, avoiding ambiguity.

Tips

- **Start with the wavetable shape.** The filter's character comes primarily from the wavetable content. Try different wavetables to find interesting spectral shapes before adjusting other parameters.
- **Sweep Frame Position for evolving textures.** Automate or modulate the Frame Position parameter to morph the filter shape over time.
- **Use Phaseless mode on transient-heavy material** (drums, percussion) to avoid the pre-ringing artifacts of linear-phase filtering.
- **Use Raw mode for sustained sounds** (pads, drones) where linear-phase clarity matters more than transient preservation.
- **The Filter Response view shows your input spectrum** as a faint amber shadow when audio is playing. Use this to visually align the cutoff with the content you want to shape.

- **Click the wavetable view** to toggle between the 3D overview (all frames) and a 2D face-on view of the current interpolated frame.

System Requirements

- **Formats:** VST3, CLAP, Standalone
- **OS:** Linux (other platforms may work but are untested)
- **CPU:** x86_64 or Apple Silicon (SIMD via Rust's portable `std::simd`)
- **DAW:** Any VST3 or CLAP compatible host (tested with Bitwig Studio)

Building from Source

Requires nightly Rust (for portable SIMD).

```
# Plugin bundles (VST3 + CLAP)
cargo nih-plug bundle wavetable-filter --release

# Standalone binary
cargo build --bin wavetable-filter --release
```